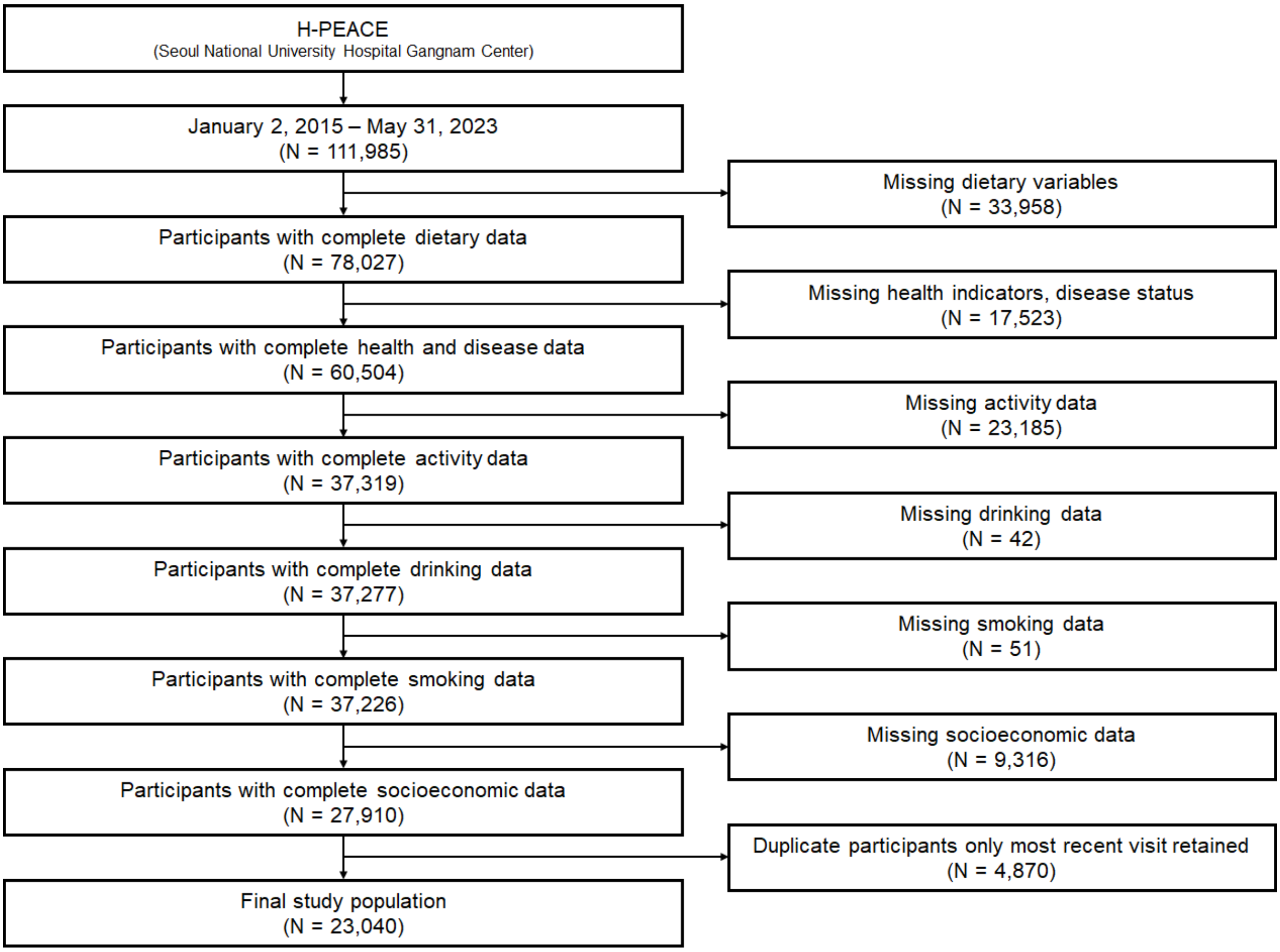
< 한국인 성별·연령별 식습관-건강 네트워크 분석 >

1. Materials and Methods

1.1. Data

1.1.1. Study Design and Population

- H-PEACE 건강검진 데이터: 총 23,040 명
- 성별·연령별 분포: 남성 12,294 명, 여성 10,746 명



1.2. 변수 정의 및 측정

1.2.1. **식품군 섭취 질** (**12 개 식품군**)

- Grain, Protein, Vegetables, Fruits, Dairy, Sweet Food, Fried, High Fat Meat, Processed, Salty Food, SSB, Add-Salt
    - 측정 척도: Poor(1 점), Intermediate(3 점), Ideal(5 점)

1.2.2. **건강 결과 변수**

- **7 가지 만성질환**: 고혈압, 당뇨병, 고지혈증, 협심증/심근경색, 뇌졸중, 만성신장질환, 비만
    - 측정 척도: 질병의 유무 (0=없음, 1=있음)
- **5 가지 대사증후군 구성요소**: 복부비만, 혈압상승, 혈당이상, 고중성지방, HDL-C 저하
    - MetS 기준의 정상/이상 (0=정상, 1=이상)
- **11 개 생체지표**: SBP, DBP, Weight, BMI, WC, FBS, TG, HBA1C, HDL-C, LDL-C, eGFR
    - 정량적 측정값 (연속변수)

1.2.3. **생활습관 변수**

- 신체활동(PA), 음주(Alcohol), 흡연(Smoking)
    - 범주형 변수

1.3. 네트워크 분석 방법론

1.3.1. **식습관 네트워크**

- **노드**: 식품군
  - 중심성 지표: Degree, Betweenness 계산
- **엣지**
  - Poor diet network: Poor 하게 섭취한 식품군끼리의 동시 발생 계수
  - Ideal diet network: Ideal 하게 섭취한 식품군끼리의 동시 발생 계수
- 네트워크 임계값: 상관계수 ≥0.1, p<0.05

1.3.2. **식습관 네트워크**

- **노드**: 식품군 또는 건강 지표
  - 중심성 지표: Degree, Betweenness 계산
- **엣지**: 식품군 간 또는 식품군과 건강 지표 간 상관계수
  - 순서형-이진변수: Point-biserial 상관계수 (식품군 점수 - 질병/MetS 유무)
  - 순서형-연속변수: Spearman 상관계수 (식품군 점수 - 생체지표 수치)
  - 이진-이진변수: Chi-square 기반 Phi 계수 (질병-MetS 간 연관성)
- 네트워크 임계값: 상관계수 ≥0.1, p<0.05

2. 결과

Table 1. Baseline characteristics

Variable	Total	MetS	Non-MetS	p-value
Age	48.62±11.33	53.35±10.80	46.98±11.04	0.000
Male	12294 (53.4%)	4425 (74.5%)	7869 (46.0%)	0.000
Education Level				**0.000**
대학원이상	7459 (32.4%)	1866 (31.4%)	5593 (32.7%)	
대학교졸업	12814 (55.6%)	3092 (52.1%)	9722 (56.9%)	
고졸~대학교재학,중퇴	2417 (10.5%)	801 (13.5%)	1616 (9.4%)	
중졸~고등학교재학,중퇴	201 (0.9%)	99 (1.7%)	102 (0.6%)	
중졸미만	149 (0.6%)	81 (1.4%)	68 (0.4%)	
Marital Status				**0.000**
사별	385 (1.7%)	164 (2.8%)	221 (1.3%)	
별거	57 (0.2%)	22 (0.4%)	35 (0.2%)	
이혼	467 (2.0%)	127 (2.1%)	340 (2.0%)	
기혼	19102 (82.9%)	5238 (88.2%)	13864 (81.1%)	
미혼	3029 (13.1%)	388 (6.5%)	2641 (15.4%)	
Household Income				**0.099**
2000 만원이상	4932 (21.4%)	1257 (21.2%)	3675 (21.5%)	
1500~2000 만원미만	2601 (11.3%)	706 (11.9%)	1895 (11.1%)	
1000~1500 만원미만	4829 (21.0%)	1220 (20.5%)	3609 (21.1%)	
800~1000 만원미만	3458 (15.0%)	847 (14.3%)	2611 (15.3%)	
500~800 만원미만	3556 (15.4%)	919 (15.5%)	2637 (15.4%)	
300~500 만원미만	2608 (11.3%)	691 (11.6%)	1917 (11.2%)	
300 만원미만	1056 (4.6%)	299 (5.0%)	757 (4.4%)	
BMI category				**0.000**
>=30	876 (3.8%)	689 (11.6%)	187 (1.1%)	
25<=x<30	5969 (25.9%)	3157 (53.2%)	2812 (16.4%)	
23<=x<25	5265 (22.9%)	1335 (22.5%)	3930 (23.0%)	
18.5<=x<23	9490 (41.2%)	744 (12.5%)	8746 (51.1%)	
<18.5	1440 (6.2%)	14 (0.2%)	1426 (8.3%)	
WC (M>=90, F>=85)	**7449 (32.3%)**	**4442 (74.8%)**	**3007 (17.6%)**	**0.000**
Anthropometric Measures				
Height (cm)	167.45±8.43	169.56±8.58	166.71±8.25	0.000
Weight (kg)	65.96±13.54	75.94±13.21	62.50±11.83	0.000
BMI(kg/m2)	23.35±3.46	26.27±3.24	22.34±2.91	0.000
Waist circumference (cm)	84.17±9.38	92.43±7.76	81.31±8.11	0.000
Disease status & Medication				
Hypertension	4050 (17.6%)	2728 (45.9%)	1322 (7.7%)	0.000
Antihypertensive Medication	3593 (15.6%)	2530 (42.6%)	1063 (6.2%)	0.000
Diabetes Mellitus	1497 (6.5%)	1034 (17.4%)	463 (2.7%)	0.000
Antidiabetic Medication	1244 (5.4%)	905 (15.2%)	339 (2.0%)	0.000
Hyperlipidemia	4848 (21.0%)	2739 (46.1%)	2109 (12.3%)	0.000
Statin Medication	3761 (16.3%)	2380 (40.1%)	1381 (8.1%)	0.000
Angina/Myocardial Infarction	471 (2.0%)	259 (4.4%)	212 (1.2%)	0.000
Stroke	136 (0.6%)	63 (1.1%)	73 (0.4%)	0.000
Chronic kidney disease (eGFR<60)	288 (1.2%)	170 (2.9%)	118 (0.7%)	0.000
Clinical Measures				
Systolic blood pressure (mmHg)	117.52±13.94	126.12±13.57	114.53±12.78	0.000
Diastolic Blood Pressure (mmHg)	75.70±10.18	81.75±9.73	73.60±9.47	0.000
Total Cholesterol (mg/dL)	202.68±39.16	195.86±44.19	205.05±36.96	0.000
Triglycerides (mg/dL)	87.00 (62.00-129.00)	145.00 (97.00-196.00)	77.00 (57.00-106.00)	0.000
LDL-C (mg/dL)	127.78±35.61	123.70±39.39	129.20±34.09	0.000
HDL-C (mg/dL)	55.19±13.34	46.72±10.67	58.13±12.91	0.000
Non-HDL-C (mg/dL)	147.49±38.78	149.14±44.81	146.91±36.44	0.001
Fasting glucose (mg/dL)	98.97±17.70	111.90±22.65	94.49±12.87	0.000
HbA1c (%)	5.57±0.59	5.92±0.78	5.45±0.45	0.000
eGFR (mL/min/1.73m2)	93.93±14.23	89.25±14.52	95.56±13.77	0.000
Obese				**0.000**
Obese	6845 (29.7%)	3846 (64.8%)	2999 (17.5%)	
Overweight	5265 (22.9%)	1335 (22.5%)	3930 (23.0%)	
Normal	10930 (47.4%)	758 (12.8%)	10172 (59.5%)	
Physical activity				**0.951**
Active	2545 (11.0%)	651 (11.0%)	1894 (11.1%)	
Minimally active	20466 (88.8%)	5281 (88.9%)	15185 (88.8%)	
Inactive	29 (0.1%)	7 (0.1%)	22 (0.1%)	
Alcohol Consumption				**0.000**

Heavy drinker	1834 (8.0%)	801 (13.5%)	1033 (6.0%)	
Mild to moderate drinker	12916 (56.1%)	3219 (54.2%)	9697 (56.7%)	
None	8290 (36.0%)	1919 (32.3%)	6371 (37.3%)	
Current smoking	8815 (38.3%)	3280 (55.2%)	5535 (32.4%)	0.000
Grain Products				**0.000**
Ideal	6299 (27.3%)	1791 (30.2%)	4508 (26.4%)	
Intermediate	13068 (56.7%)	3121 (52.6%)	9947 (58.2%)	
Poor	3673 (15.9%)	1027 (17.3%)	2646 (15.5%)	
Protein Foods				**0.865**
Ideal	6249 (27.1%)	1622 (27.3%)	4627 (27.1%)	
Intermediate	11166 (48.5%)	2881 (48.5%)	8285 (48.4%)	
Poor	5625 (24.4%)	1436 (24.2%)	4189 (24.5%)	
Vegetables				**0.649**
Ideal	4741 (20.6%)	1218 (20.5%)	3523 (20.6%)	
Intermediate	9981 (43.3%)	2548 (42.9%)	7433 (43.5%)	
Poor	8318 (36.1%)	2173 (36.6%)	6145 (35.9%)	
Dairy Products				**0.000**
Ideal	6018 (26.1%)	1483 (25.0%)	4535 (26.5%)	
Intermediate	7661 (33.3%)	1875 (31.6%)	5786 (33.8%)	
Poor	9361 (40.6%)	2581 (43.5%)	6780 (39.6%)	
Fruits				**0.293**
Ideal	7876 (34.2%)	2021 (34.0%)	5855 (34.2%)	
Intermediate	8878 (38.5%)	2253 (37.9%)	6625 (38.7%)	
Poor	6286 (27.3%)	1665 (28.0%)	4621 (27.0%)	
Fried Foods				**0.830**
Ideal	15731 (68.3%)	4048 (68.2%)	11683 (68.3%)	
Intermediate	6072 (26.4%)	1563 (26.3%)	4509 (26.4%)	
Poor	1237 (5.4%)	328 (5.5%)	909 (5.3%)	
High Fat Meat				**0.000**
Ideal	13303 (57.7%)	3213 (54.1%)	10090 (59.0%)	
Intermediate	8294 (36.0%)	2328 (39.2%)	5966 (34.9%)	
Poor	1443 (6.3%)	398 (6.7%)	1045 (6.1%)	
Processed Foods				**0.077**
Ideal	14320 (62.2%)	3760 (63.3%)	10560 (61.8%)	
Intermediate	6945 (30.1%)	1748 (29.4%)	5197 (30.4%)	
Poor	1775 (7.7%)	431 (7.3%)	1344 (7.9%)	
Sugar-Sweetened Beverages				**0.005**
Ideal	16500 (71.6%)	4158 (70.0%)	12342 (72.2%)	
Intermediate	4196 (18.2%)	1133 (19.1%)	3063 (17.9%)	
Poor	2344 (10.2%)	648 (10.9%)	1696 (9.9%)	
Additional Salt Use				**0.000**
Ideal	13004 (56.4%)	3199 (53.9%)	9805 (57.3%)	
Intermediate	8903 (38.6%)	2431 (40.9%)	6472 (37.8%)	
Poor	1133 (4.9%)	309 (5.2%)	824 (4.8%)	
Salty Food Consumption				**0.000**
Ideal	9401 (40.8%)	2014 (33.9%)	7387 (43.2%)	
Intermediate	11246 (48.8%)	3137 (52.8%)	8109 (47.4%)	
Poor	2393 (10.4%)	788 (13.3%)	1605 (9.4%)	
Sweet Food Consumption				**0.000**
Ideal	3170 (13.8%)	1048 (17.6%)	2122 (12.4%)	
Intermediate	15497 (67.3%)	4044 (68.1%)	11453 (67.0%)	
Poor	4373 (19.0%)	847 (14.3%)	3526 (20.6%)	
Metabolic syndrome				
Increased waist circumference	7449 (32.3%)	4442 (74.8%)	3007 (17.6%)	0.000
Elevated blood pressure	8240 (35.8%)	4741 (79.8%)	3499 (20.5%)	0.000
Impaired fasting glucose	8673 (37.6%)	4810 (81.0%)	3863 (22.6%)	0.000
Elevated triglycerides	7018 (30.5%)	4590 (77.3%)	2428 (14.2%)	0.000
Decreased HDL-C	3988 (17.3%)	2341 (39.4%)	1647 (9.6%)	0.000
Metabolic syndrome	5939 (25.8%)	5939 (100.0%)	0 (0.0%)	0.000

2.1. 전체 인구집단 네트워크 분석 (n=23,040)

2.1.1. 식습관 네트워크 분석

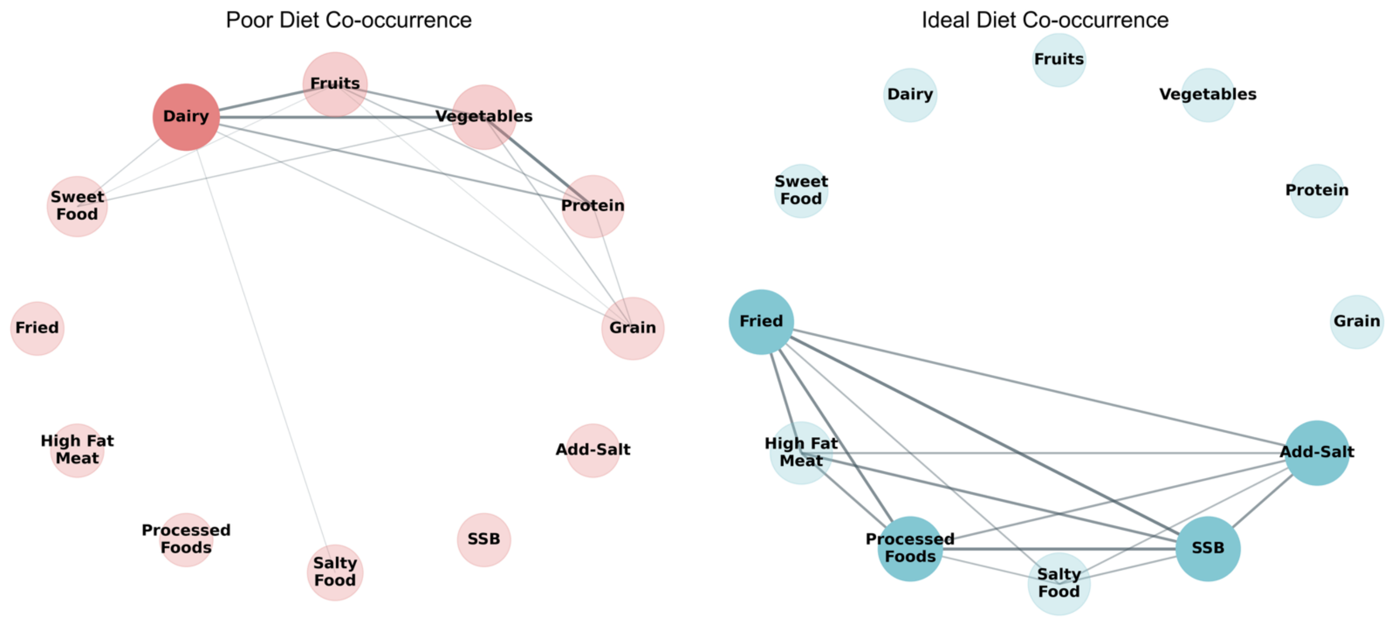


Figure 1 Total Diet Co-occurrence network

- **[Figure 1(A)] Poor Diet Co-occurrence 네트워크**
  - **Degree centrality** 상위 식품군 (크기)
    : Dairy(0.545) > Vegetables(0.455) = Fruits(0.455) > Grain(0.364) = Protein(0.364)
  - **Betweenness centrality** 상위 식품군 (진하기)
    : Dairy(0.103) > Vegetables(0.012) = Fruits(0.012)
  - **Co-occurrence** (동시에 poor 하게 섭취되는 식품군 조합 빈도)
    * Protein + Vegetables: 4,147 명
    * Vegetables + Dairy: 3,912 명
    * Fruits + Dairy: 3,691 명
  - **결론: 핵심요소 = Dairy (유제품)**

- **[Figure 1(B)] Ideal Diet Co-occurrence 네트워크**
  - **Degree centrality** 상위 식품군 (크기)
    : Fried(0.455) = Processed Foods(0.455) = SSB(0.455) = Add-Salt(0.455) > High Fat Meat(0.364)
  - **Betweenness centrality** 상위 식품군 (진하기)
    : Fried(0.005) = Processed Foods(0.005) = SSB(0.005)
  - Co-occurrence (동시에 ideal 하게 섭취되는 식품군 조합 빈도)
    * Fried + SSB: 12,289 명
    * Processed Foods + SSB: 11,882 명
    * Fried + Processed Foods: 11,453 명
  - **결론: 핵심 요소 = Fried/SSB/Processed Foods**

2.1.2. 건강지표별 네트워크 분석

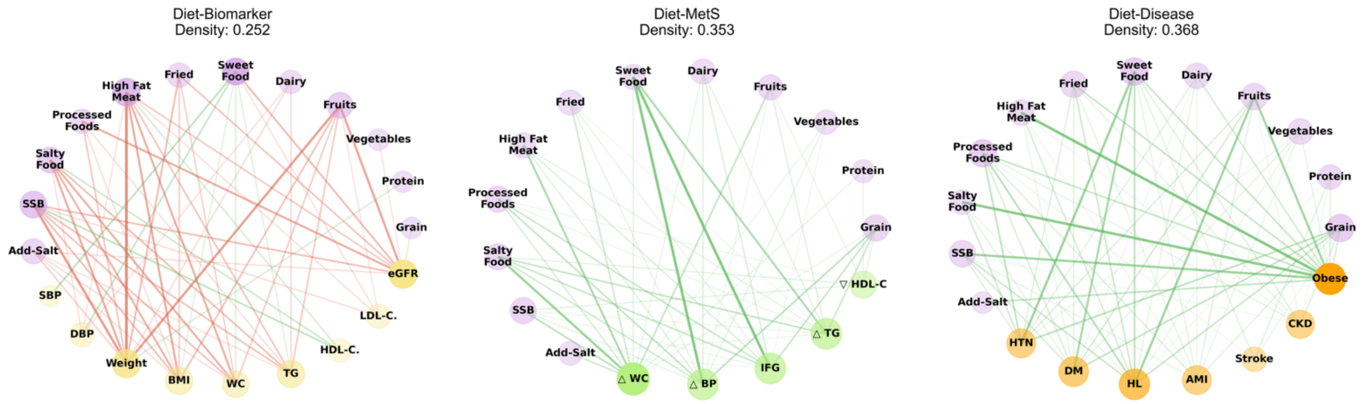


Figure 2 Diet – Health Correlation Network

- **[Figure 2 (A)] Diet-Biomarker 네트워크 (밀도: 0.252)**
  - **Degree centrality** 상위 건강지표: Weight(0.450) > eGFR(0.450) > BMI(0.400)
  - **부정적 상관관계** 우세: 건강한 식품군 섭취가 위험 생체지표 감소와 연관

- **[Figure 2 (B)] Diet-MetS 네트워크 (밀도: 0.353)**
  - **Degree centrality** 상위 건강지표: △WC(0.750) > △BP(0.625) > IFG(0.625)
  - **긍정적 상관관계** 우세: MetS 위험 증가 시 식습관 개선 패턴

- **[Figure 2 (C)] Diet-Disease 네트워크 (밀도: 0.368)**
  - **Degree centrality** 상위 건강지표: Obese(0.667) > HL(0.611) > DM(0.556)
  - **긍정적 상관관계** 우세: 질병 보유 시 식습관 개선 패턴

- **결론**
  - **생체지표만 안 좋은 초기**에는 **"안 좋은 식습관 → 안 좋은 생체지표"** 패턴
  - **질병/MetS 이 이미 유발된 후**에는 오히려 **수진자가 식습관을 관리**한 것과 같은 패턴

2.2. 성별 간 네트워크 차이 분석

2.2.1. 식습관 네트워크 분석

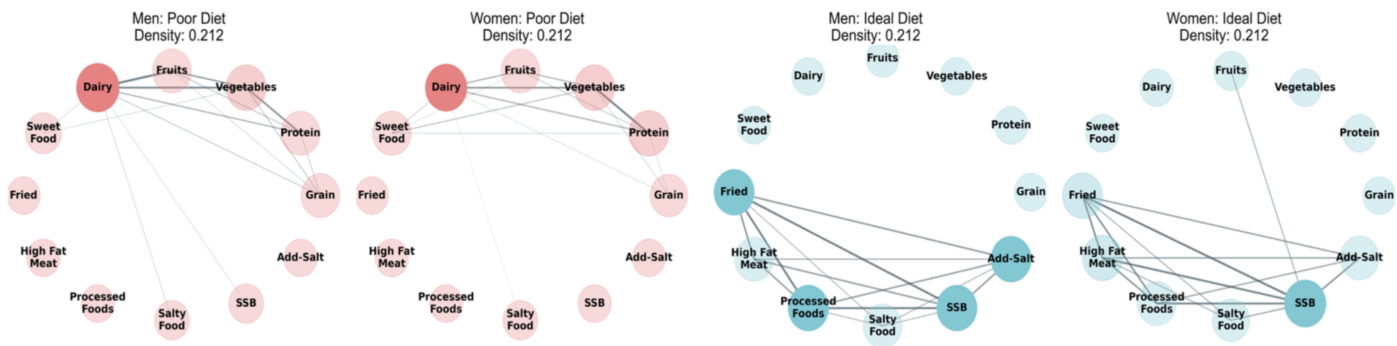


Figure 3 Diet Cooccurrence Gender Comparison

- **[Figure 3 (A)] Poor Diet 네트워크**
  - 남성: Dairy(0.64) > Vegetables(0.45) > Grain(0.36)
  - 여성: Dairy(0.55) > Protein(0.45) = Vegetables(0.45)
  - 네트워크 밀도: 남녀 모두 0.212 로 동일
  - 연결 수: 남녀 모두 14 개
  - **결론: 남녀 모두 Poor Diet 에 대한 핵심 요소 및 주요 지표는 비슷함.**

- **[Figure 3 (B)] Ideal Diet 네트워크**
  - 남성: Fried(0.45) = Processed Foods(0.45) = SSB(0.45)
  - 여성: SSB(0.55) > Fried(0.45) = High Fat Meat(0.45)
  - 네트워크 밀도: 남녀 모두 0.212 로 동일
  - 연결 수: 남녀 모두 14 개
  - **결론: 남녀 모두 Ideal Diet 에 대한 핵심 요소 및 주요 지표는 비슷함.**

2.2.2. 건강지표별 네트워크 분석

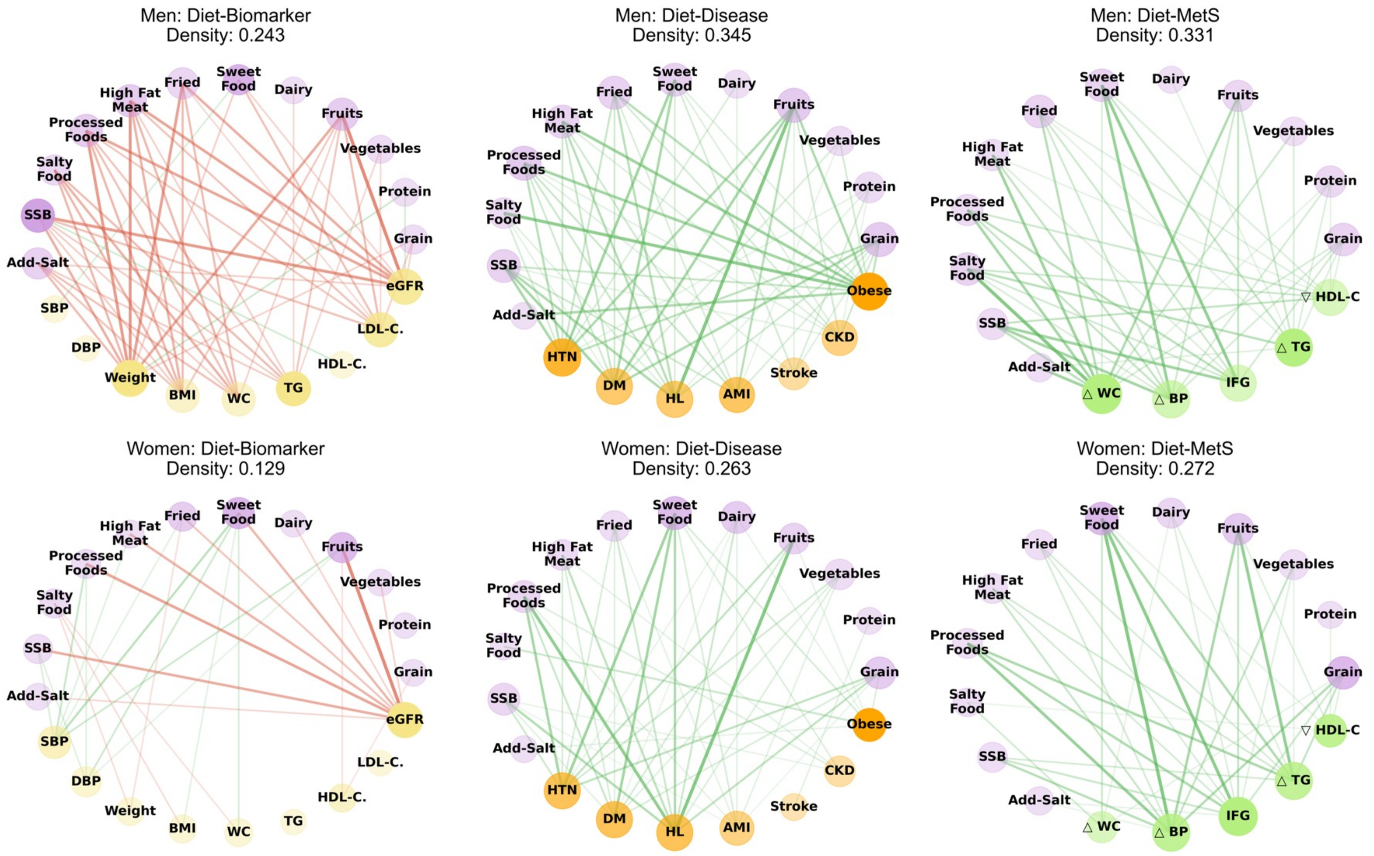


Figure 4 Diet-Health Networks: Gender Comparison

- **네트워크 밀도 비교**
  - Figure 4 (A) **질병**: 남성 0.345 > 여성 0.263 (**남성이 31% 높음**)
  - Figure 4 (B) **MetS**: 남성 0.331 > 여성 0.272 (**남성이 22% 높음**)
  - Figure 4 (C) **생체지표**: 남성 0.243 > 여성 0.129 (**남성이 88% 높음**)
  - **결론: 남성이 여성보다 식습관이 건강지표와 더 큰 상관성을 보임**

- **식습관과 건강지표 간의 상관성 (p<0.05)**
  - 총 **16 개** 변수에서 Cohen's d > 0.8
  - 주요 **남성** 우세 변수: **TG, Weight, BMI, WC, LDL-C, Protein, Add-Salt**
  - 주요 **여성** 우세 변수: **SBP, DBP, Dairy**

2.3. 성별 및 연령별 네트워크 차이 분석

2.3.1. 식습관 네트워크 분석

- 남성 연령별 네트워크 구조 변화

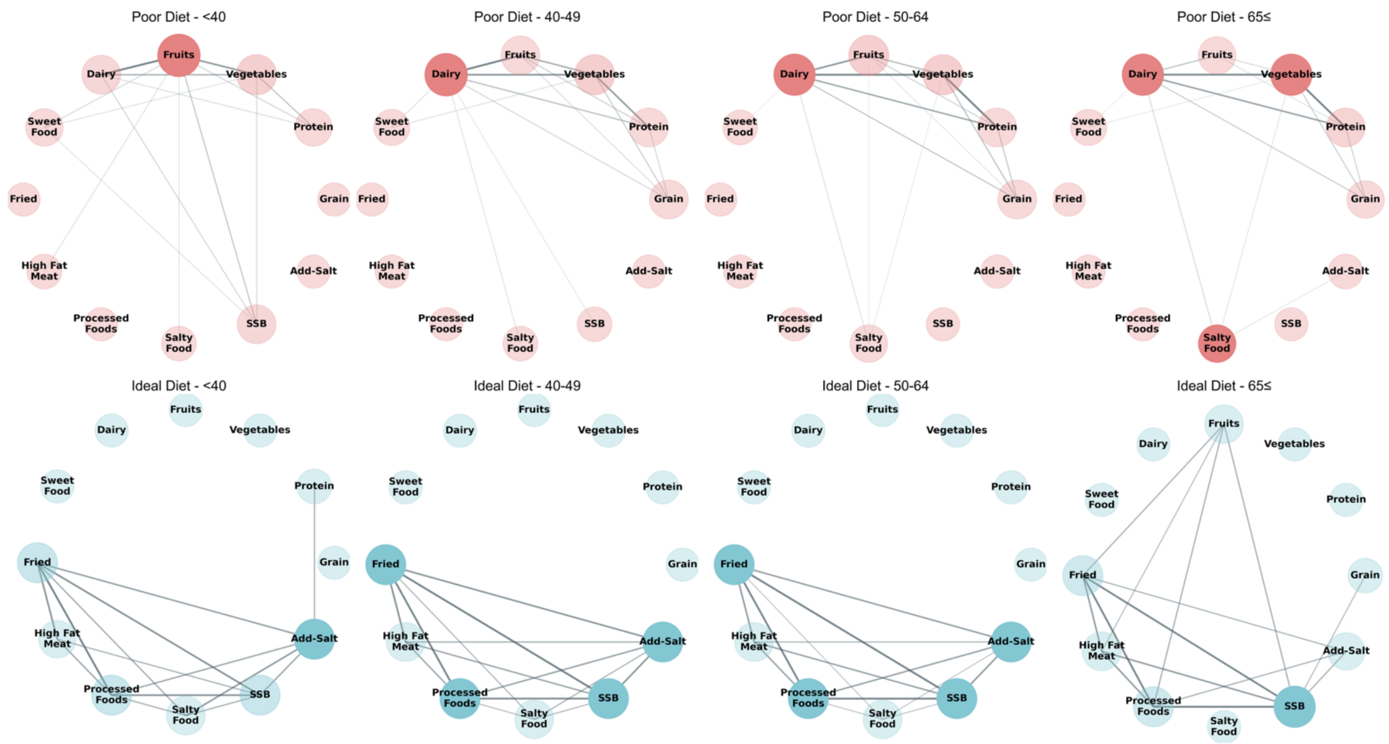


Figure 5 Men Diet Age Progression

- **[Figure 5 (A)] Poor Diet 네트워크**
  * **40 세 미만**: **Fruits(0.64)** > **Vegetables(0.45)** > Dairy(0.36)
  * **40-49 세**: **Dairy(0.64)** > **Vegetables(0.45)** > Grain(0.36)
  * **50-64 세**: **Dairy(0.55)** > **Vegetables(0.45)** = Fruits(0.45)
  * **65 세 이상**: **Vegetables(0.55)** = **Dairy(0.55)** > Protein(0.36)

- **[Figure 5 (B)] Ideal Diet 네트워크**
  * **40 세 미만**: **Fried(0.45) = Processed Foods(0.45) = SSB(0.45)**
  * **40-49 세**: **Fried(0.45) = Processed Foods(0.45) = SSB(0.45)**
  * **50-64 세**: **Fried(0.45) = Processed Foods(0.45) = SSB(0.45)**
  * **65 세 이상**: **SSB(0.55)** > Fried(0.45) = Processed Foods(0.45)

• **여성 연령별 네트워크 구조 변화**

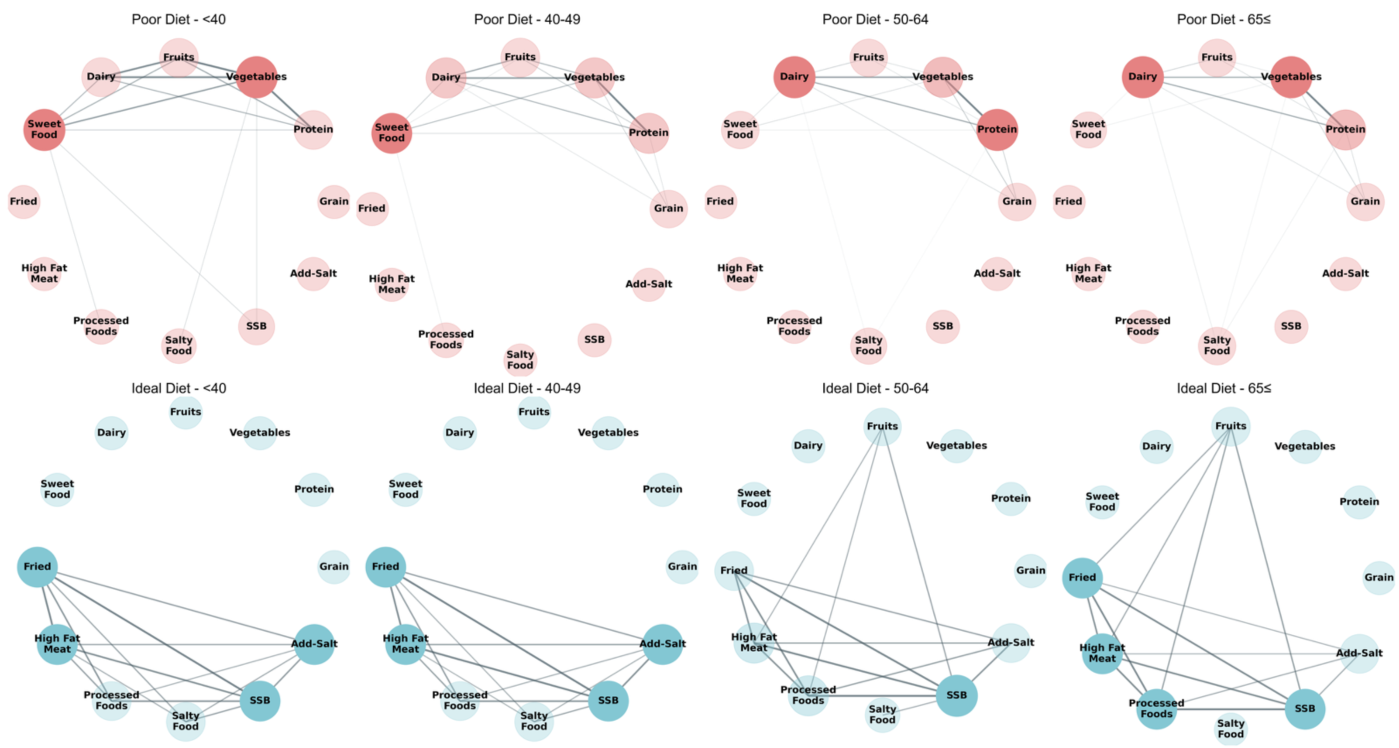


Figure 6 Women Diet Age Progression

- **[Figure 6 (A)] Poor Diet 네트워크**
  * **40 세 미만: Vegetables(0.55)** = **Sweet Food(0.55)** > Protein(0.36)
  * **40-49 세: Protein(0.45) = Vegetables(0.45) = Dairy(0.45)**
  * **50-64 세: Protein(0.55) = Dairy(0.55)** > Vegetables(0.45)
  * **65 세 이상: Vegetables(0.55) = Dairy(0.55)** > Protein(0.45)

- **[Figure 6 (B)] Ideal Diet 네트워크**
  * **40 세 미만: Fried(0.45) = High Fat Meat(0.45) = SSB(0.45)**
  * **40-49 세: Fried(0.45) = High Fat Meat(0.45) = SSB(0.45)**
  * **50-64 세: SSB(0.55)** > High Fat Meat(0.45) = Processed Foods(0.45)
  * **65 세 이상: Fried(0.45) = High Fat Meat(0.45) = Processed Foods(0.45)**

>> 생애주기별 맞춤형 식습관 중재 전략

연령대	남성	여성
40 세 미만	**과일** 섭취 증진	**채소** 섭취 + **단맛** 조절
40-64 세 (중장년)	**유제품** 섭취 강화	**단백질-유제품** 균형 섭취
65 세 이상 (노년)	채소와 유제품의 균형적 섭취 + **음료** 섭취 특별 관리	**채소**와 **유제품**의 균형적 섭취

2.3.2. 건강지표별 네트워크 분석

- 남성 연령별 네트워크 구조 변화

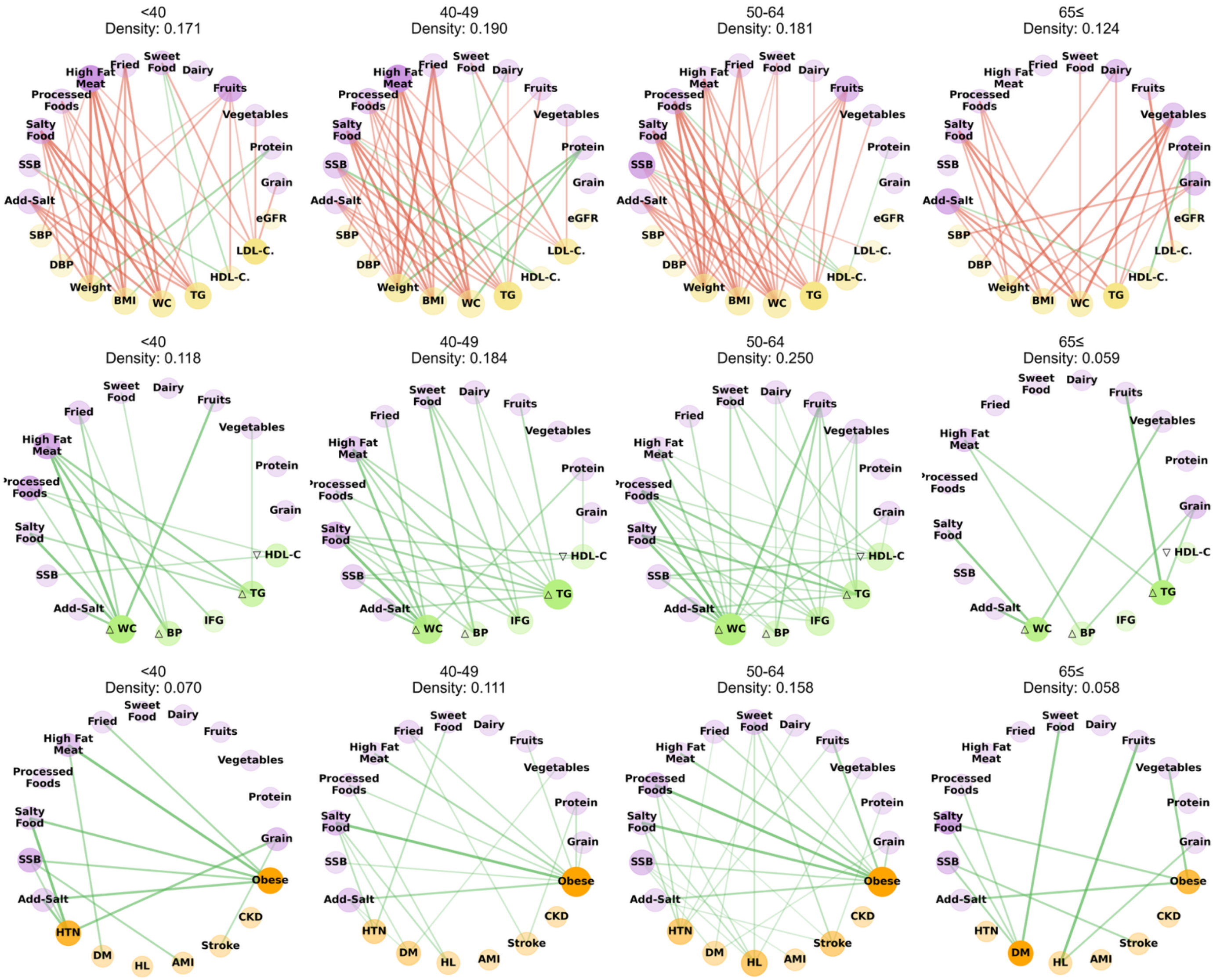


Figure 7 Men Health-Diet Age progression

- **[Figure 7 (A)] Biomarker** 네트워크 (밀도: 0.171 → **0.190** → 0.181 → 0.124):
  * **40 세 미만**: **Weight** 중심(0.300), 체형 관련 시작
  * **40-49 세**: **Weight 최고점(0.400)**, 체중 관리 핵심
  * **50-64 세**: **Weight** 중심(0.400), 체중 관리 중요
  * **65 세 이상**: **Weight 급감**(0.250)

- **[Figure 7 (B)] MetS** 네트워크 (밀도: 0.118 → 0.184 → **0.250** → 0.059):
  * **40 세 미만**: **△WC** (0.375)
  * **40-49 세**: **△TG** 증가(0.500)
  * **50-64 세**: **△WC 최고점(0.625)**, 최대 네트워크 밀도
  * **65 세 이상**: **△WC 급감**(0.188)

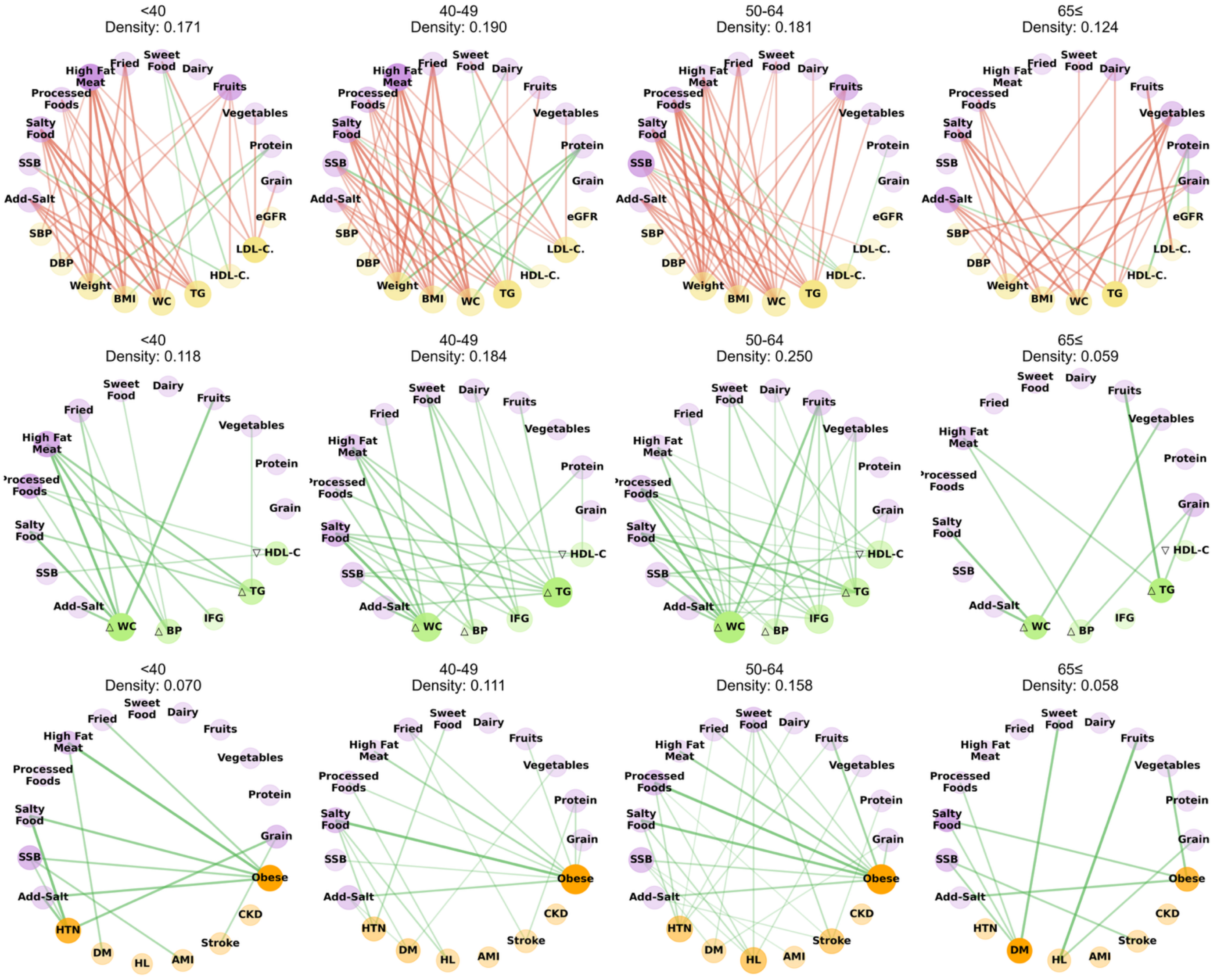


Figure 8 Men Health-Diet Age progression

- **[Figure 7 (C)] Disease** 네트워크 (밀도: 0.070 → 0.111 → **0.158** → 0.058):
  * **40 세 미만**: **Obese** (0.278), 식품군과 약한 연결
  * **40-49 세**: **Obese** (0.500), 연결 강도 증가
  * **50-64 세**: **Obese 최고점**(0.500), 최대 네트워크 밀도
  * **65 세 이상**: **DM 중심 전환**(0.222), **연결 감소**

- **결론**
  * **50 대 위험**
  * **비만 관련 지표들** 지속 핵심
  * 질병 경로: 비만→당뇨
  * **중재전략: 50 대 전후 집중 관리, 복부둘레 관련 식습관 연결고리 차단 필요**

• **여성 연령별 네트워크 구조 변화**

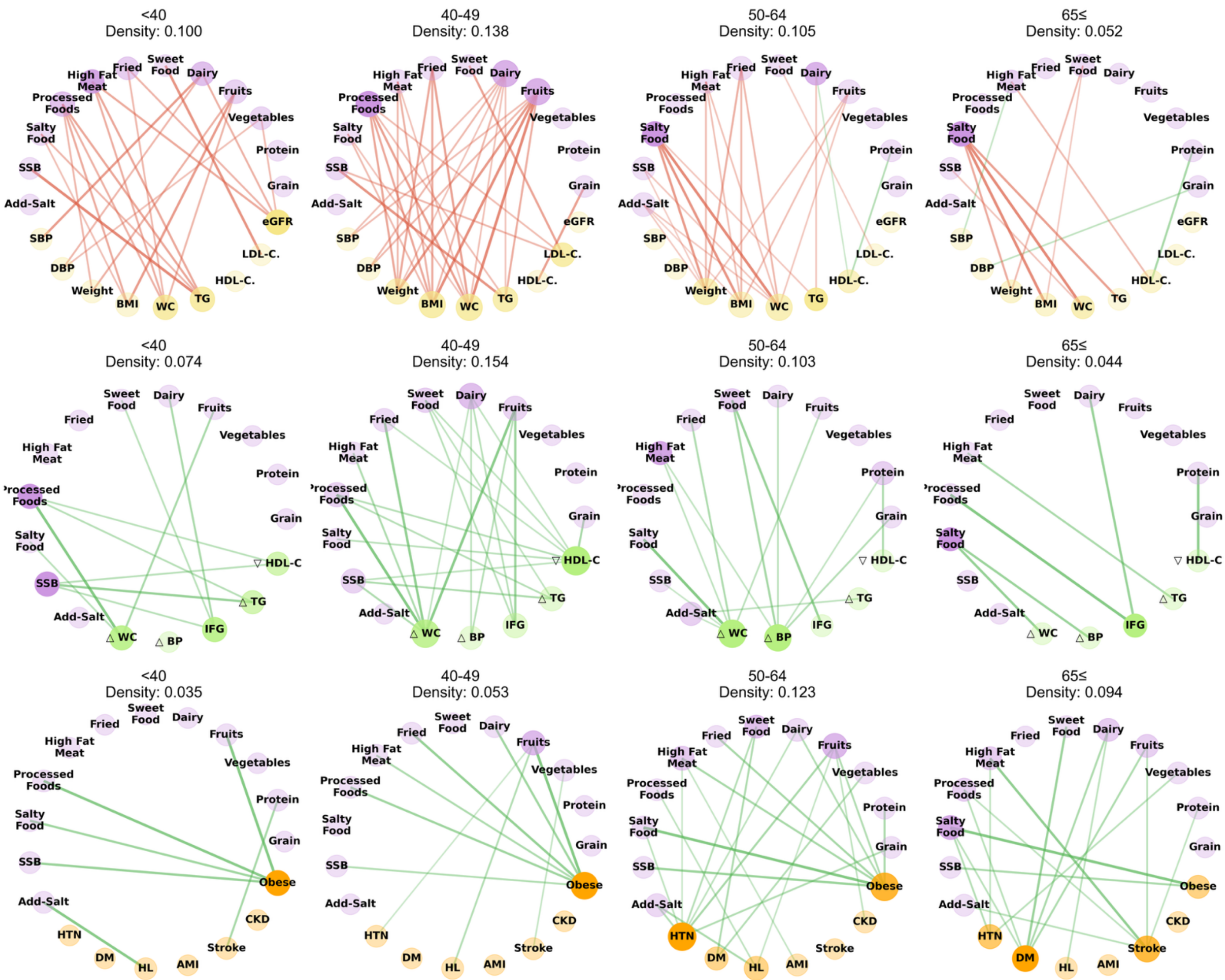


Figure 9 Women Health-Diet Age progression

- **[Figure 9 (A)]** **Biomarker** 네트워크 (밀도: 0.100 → **0.138** → 0.105 → 0.052):
  * **40 세 미만**: **WC 중심**(0.200)
  * **40-49 세: BMI 최고점**(0.300), 체중 관리 중요
  * **50-64 세**: Weight 중심(0.300)
  * **65 세 이상**: Weight 감소(0.100)

- **[Figure 9 (B)]** **MetS** 네트워크 (밀도: 0.074 → **0.154** → 0.103 → 0.044):
  * **40 세 미만**: **△WC** 중심(0.188)
  * **40-49 세**: **△WC, ▽HDL-C** 최고점(0.438)
  * **50-64 세**: **△WC 감소(0.375), 남성과 상반된** 패턴
  * **65 세 이상**: **IFG** (0.125), 약한 상관성

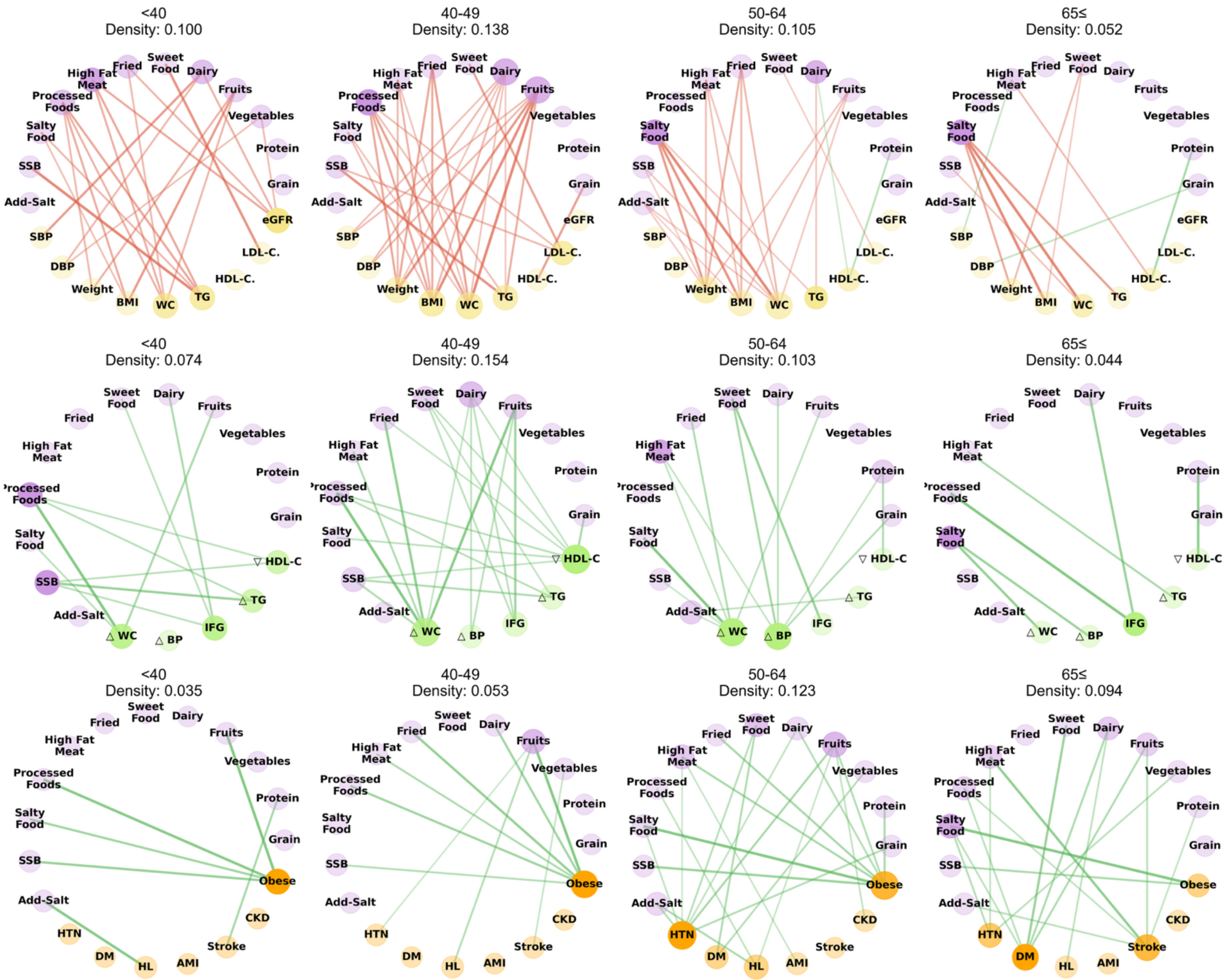


Figure 10 Women Health-Diet Age progression

- **[Figure 9 (C)] Disease** 네트워크 (밀도: 0.035 → 0.053 → **0.123** → 0.094):
  * **40 세 미만**: **Obese** (0.222), 남성보다 낮은 초기값
  * **40-49 세**: **Obese** 증가(0.333), **중년 비만** 위험
  * **50-64 세**: **HTN** 최고점(0.389), **고혈압 중심 전환**
  * **65 세 이상**: **DM** 전환(0.278), **남성보다 높은 당뇨** 연결

- **결론**
  * **40 대 위험**
  * **다원화된** 관리 필요 (IFG↔△WC 교대)
  * **중재전략: 40 대 조기 개입, 혈당-혈압-콜레스테롤 함께 모니터링 필요**

Supplementary Table 1. Factor loadings of dietary patterns identified

Variables	Definition	Feature
Grain Products	How often do you eat grains (rice, noodle, bread, rice cake, bread, cereal, and (sweet) potatoes)?	• Ideal: 3 times/day • Intermediate: 1–2 time(s)/day • Poor: Less than 6 times/week
Protein Foods	How often do you eat one of dishes like meat, fish, tofu, or eggs?	• Ideal: More than 2 times/day • Intermediate: Once a day • Poor: 3–6 times/week or Less than 2 times/ week or hardly
Vegetables	How often do you eat vegetables (raw vegetables, cooked potherb, and salad)?	• Ideal: More than 2 times/day • Intermediate: Once a day • Poor: 3–6 times/week or Less than 2 times/ week or hardly
Dairy Products	How often do you drink milk and dairy products? (1 time: milk 200mL or yogurt drink 150mL or yogurt 100g)	• Ideal: 1–2 time(s)/day • Intermediate: More than 3 times/day or 3–6 times/week • Poor: Less than 2 times/ week or hardly
Fruits	How often do you eat fruits?	• Ideal: Everyday • Intermediate: 3–6 times/week • Poor: Less than 2 times/week or hardly
Fried Foods	How often do you eat fried food?	• Ideal: Less than 1–2/week or hardly • Intermediate: 3–6 times/week • Poor: More than 2 times/day or Once a day
High Fat Meat	How often do you eat fat rich meat (pork fatback, rib, and eel)?	• Ideal: Less than 1–2/week or hardly • Intermediate: 3–6 times/week • Poor: More than 2 times/day or Once a day
Processed Foods	How often do you eat instant and processed food (instant noodle, ham, and canned food)?	• Ideal: Less than 1–2/week or hardly • Intermediate: 3–6 times/week • Poor: More than 2 times/day or Once a day
Sugar-Sweetened Beverages	How often do you drink juice, soft, or instant drinks? (1 glass: 200mL)	• Ideal: Less than 1–2 glass(es)/week or hardly • Intermediate: 3–6 glasses/week • Poor: More than 2 glasses/day or 1 glass/day
Additional Salt Use	Do you usually add more salt or soy sauce to your food?	• Ideal: Never • Intermediate: Sometimes • Poor: Often
Salty Food Consumption	Do you usually eat salty food like fermented fish, pickled vegetable or soup?	• Ideal: NO • Intermediate: Sometimes • Poor: YES
Sweet Food Consumption	Do you usually eat sweets (ice cream, cake, and cookies)?	• Ideal: Never • Intermediate: Sometimes (less than 2/week) • Poor: Often (more than 3/week)

- 식품군 간 상관성 (Spearman 계수)
  - 정적 상관관계 (r>0.3)
    * Protein ↔ Vegetables: r=0.475
    * Fried ↔ High Fat Meat: r=0.363
    * Fried ↔ Processed Foods: r=0.334
    * Processed Foods ↔ SSB: r=0.334
    * Salty Food ↔ Add-Salt: r=0.350
  - 부적 상관관계 (r<-0.3): 없음
  - 독립적 관계 (-0.1<r<0.1): 31 개 식품군 쌍이 독립적 섭취 패턴